The Assessment Question Paper and Assessment Answer Booklet must not be removed from the assessment room. Both must be handed in at the end of the assessment.

SIM Global Education

WRITTEN TEST
CONTINOUS ASSESSMENT (MAIN)

(DIPLOMA IN INFORMATION TECHNOLOGY)

Date	**:**	**22 October 2025**
Module	**:**	**ITSD001 Business Statistics with Python**
Time	**:**	**10:00 – 11:30**
Duration	**:**	**1 hour 30 minutes**
Total no. of pages (including this cover page)	**:**	**5 + 13 pages of Formula sheet/Financial Tables**

INSTRUCTIONS TO CANDIDATES:

1. This assessment carries a total of 100 marks. Answer **ALL** questions.
2. Answer **ALL** questions in the answer booklets provided. Fill in the cover page carefully. **You will receive a zero (0) score if your name is filled incorrectly**.
3. All answers must be clearly written. Begin each question on a fresh page.
4. The use of non-programmable calculators **IS** permitted in this assessment.
5. This is a closed-book assessment. No unauthorised materials are allowed in the assessment venue.
6. A **THIRTEEN (13)–page** Formula sheet and financial tables will be given separately.

At the end of the assessment:
Please ensure that you have written your name on each answer booklet used: failure to do so will mean that your work cannot be identified. If you have used more than one answer booklet, please tie them together with the string provided.

SIM GLOBAL EDUCATION RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

SAMPLE

ANSWER ALL QUESTIONS

Question 1

(a) Mr Tan is opening an online store specializing in refurbished laptops. To understand the market price range, he visited 10 different online marketplaces and collected the prices (in $'00) of laptops with similar specifications:

12	18	18	23	29	35	40	42	47	55

Compute the mean, median, mode, range and standard deviation. (Express your answer to 2 decimal places).

(10 marks)

(b) Sharon wants to estimate the monthly household expenses on electricity. She conducted a small survey with 10 families in her neighborhood. Their monthly bills (in $) are as follows:

Amount Spent ($)	85	92	95	105	108	111	117	118	134	138

Construct a stem and leaf diagram from the above data. Use stem unit = $10 and leaf unit = $1.

(6 marks)

(c) Mr Tan also sells tablets in his store. He recorded the number of tablets sold daily over a 20-day period:

Number of tablets sold	**Number of days (Frequency)**	**Relative Frequency (%)**	**Class Mid-point**
0 up to 3	4		
3 up to 6	7		
6 up to 9	5		
9 up to 12	3		
12 up to 15	1		
Total	20		

Based on the frequency distribution table above, calculate the relative frequencies (Express your answer to 2 decimal places) and class mid-points. Construct a histogram based on the frequency distribution table provided. Label all axes clearly.

(9 marks)

SAMPLE

Question 2

(a) A technology company is hiring 10 data analyst interns from a pool of 200 applicants. Applicants are classified by education level and preferred department as shown below.

Department	Diploma Holders	Degree Holders	Total
IT	45	35	80
Marketing	25	40	65
Finance	15	40	55
Total	85	115	200

Given that the applicant is from marketing, what is the probability that the applicant is a degree holder? What is the probability that an applicant is from IT **or** the applicant is a diploma holder? (Express answers to 3 decimal places)

(8 marks)

(b) The technology company also wants to estimate monthly server maintenance costs. Historical data suggest the cost follows a normal distribution with mean $\boldsymbol{\mu}$= 1200 and standard deviation $\boldsymbol{\sigma}$ = 150, what is P(1200<X<1300)? If the company randomly samples 25 months, calculate the probability that the average monthly cost exceeds $1,230? (Express answer to 3 decimal places)

(8 marks)

(c) The technology company organized a team-building event featuring a lucky spin game for the new hires. Each play costs $50. If a star appears (P=1/5), the player gets $100. If a moon appears (P=1/20), the player gets $200. If a crown appears (P=1/100), the player gets $500. Otherwise, the player gets $0. What amount can a player expect to win or lose by playing the game?

(9 marks)

SAMPLE

Question 3

(a) A customer service center receives an average of 3 complaints per hour, and the number of complaints follows a Poisson distribution. Find the probability that in a particular hour the center receives:

(i) Exactly 4 complaints.

(ii) At least 2 complaints. (Correct your answers to 4 decimal places)

(10 marks)

(b) A battery factory produces rechargeable batteries, where 1 in 8 is defective**.** 16 batteries are tested at random. Find the probability that less than three of them will be defective. (Correct your answer to 4 decimal places)

(9 marks)

(c) State the level of measurement (Ratio, Ordinal or Nominal) for the variables: "Number of online purchases per month" and "Favourite smartphone brand".

(6 marks)

SAMPLE

Question 4

(a) A survey of 20 software engineers found their average weekly coding hours to be 45.3, with sample standard deviation 8.2 hours. Find the 99% confident interval of the population mean. (Express your answer to 3 decimal places)

(9 marks)

(b) Another 30 engineers were surveyed, and the mean and standard deviation remained the same. Find the 95% confident interval for the population mean hours. (Express your answer to 3 decimal places.)

(9 marks)

(c) The surveyor has made a mistake in calculating the standard deviation and the correct sample standard deviation has a smaller value of 5. How would this affect the width of the confidence interval?

(7 marks)

--- END OF PAPER ----

SAMPLE

Business Statistic with Python Formula Sheet

Numerical Descriptive Measure

Sample Variance

$$S^2 = \frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\mathrm{n} - 1}$$

Sample Standard Deviation

$$S = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \bar{X})^2}{\mathrm{n} - 1}}$$

Z-score Formula

$$Z = \frac{X - \bar{X}}{S}$$

Population Mean

$$\mu = \frac{\sum_{i=1}^{N} X_i}{N}$$

Population Variance

$$\sigma^2 = \frac{\sum_{i=1}^{n}(X_i - \mu)^2}{\mathrm{N}}$$

Population Standard Deviation

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n}(X_i - \mu)^2}{\mathrm{N}}}$$

Sample Covariance

$$cov(X, Y) = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Coefficient of Correlation

$$r = \frac{cov(X, Y)}{S_X S_Y}$$

Basic Probability

Probability of a joint event

$$P(A\ and\ B) = \frac{\text{number of outomes satisfying A and B}}{\text{total number of elementary outcomes}}$$

Marginal Probability

$$P(A\ and\ B) = P(A\ and\ B_1) + P(A\ and\ B_2)\ +\ \ldots\ P(A\ and\ B_k)$$

General Addition Rule

$$P(A\ or\ B)\ =\ P(A)\ +\ P(B) - P(A\ and\ B)$$

Conditional Probability

$$P(A|B) = \frac{\mathrm{P(A\ and\ B)}}{\mathrm{P(B)}}$$

Multiplication Rule

$$P(A\ and\ B)\ =\ P(A|B)\ P(B)$$

Bayes' Theorem

$$\mathrm{P(B_i|A)} = \frac{\mathrm{P(A|B_i)P(B_i)}}{\mathrm{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_k)P(B_k)}}$$

Discrete Probability Distributions

SAMPLE

Mean of a discrete distribution	$\mu = E(X) = \sum_{i=1}^{N} X_i P(X_i)$
Variance of a discrete random variable	$\sigma^2 = \sum_{i=1}^{N} [X_i - E(X)]^2 P(X_i)$
Standard deviation of a discrete random variable	$\sigma = \sqrt{\sigma^2}$
Covariance	$\sigma_{XY} = \sum_{i=1}^{N} [X_i - E(X)][Y_i - E(Y)] P(X_i Y_i)$
Expected value of the sum of two random variables	E(X + Y) = E(X) + E(Y)
Variance of the sum of two random variables	Var(X + Y) =$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2 + 2\sigma_{XY}$
Standard deviation of the sum of two random variables	$\sigma_{XY} = \sqrt{\sigma_{X+Y}^2}$
Portfolio Expected return	E(P) = wE(X) + (1-w)E(Y)
Portfolio Risk	$\sigma_p = \sqrt{w^2\sigma_X^2 + (1-w)^2\sigma_Y^2 + 2w(1-w)\sigma_{XY}}$
Rule of Combinations	$\binom{n}{x} = \frac{n!}{X!\,(n-X)!}$
Binomial Distribution Formula	$P(X = x \mid n, \pi) = \frac{n!}{x!\,(n-x)!} \pi^x (1-\pi)^{n-x}$
Poisson Distribution Formula	$P(X) = \frac{e^{-\lambda}\lambda^x}{X!}$

Continuous Distributions and the Normal Distributions

Normal probability density function	$f(X) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}(\frac{(X-\mu)}{\sigma})^2}$
Standardised Normal	$Z = \frac{X-\mu}{\sigma}$
Standardised normal probability density function	$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-(\frac{1}{2})Z^2}$

SAMPLE

X value from Z Value	$X = \mu + Z\sigma$
Mean of a uniform distribution	$\mu = \frac{a+b}{2}$
Standard deviation of a uniform distribution	$\sigma = \sqrt{\frac{(b-a)^2}{12}}$
Variance of a uniform distribution	$\sigma^2 = \frac{(b-a)^2}{12}$
Density function for uniform distribution	$f(x) = \begin{cases} \frac{1}{b-a} & if\ a \le X \le b \\ 0 & otherwise \end{cases}$

Sampling Distributions

Mean of a Population Distribution	$\frac{\sum X_i}{N}$
Standard Deviation of a Population Distribution	$\sigma = \sqrt{\frac{\sum (X_i - \mu)^2}{N}}$
Standard Error of the Mean	$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$
Z-Value for a Sampling Distribution	$Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}}$
Sample Proportion	$p = \frac{number\ if\ items\ in\ sample\ of\ interest}{sample\ size}$
Z-value for Proportions	$Z = \frac{p - \pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$
Finite Population Correction Factor	$\sqrt{\frac{N-n}{N-1}}$
Standard Error of the Mean for Finite Population	$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}$

SAMPLE

Standard Error of the Proportion for Finite Populations	$\sigma_{\bar{X}} = \sqrt{\frac{\pi(1-\pi)}{n}}\sqrt{\frac{N-n}{N-1}}$

Confidence Interval Estimation

Point Estimate	Point Estimate $\pm$ (Critical Value)(Standard Error)
Confidence Interval for μ, σ known	$\bar{X} \pm Z_{\alpha/2}\frac{\sigma}{\sqrt{n}}$
Confidence Interval for μ, σ unknown	$\bar{X} \pm t_{\alpha/2}\frac{S}{\sqrt{n}}$
Confidence Interval for Population Proportion π	$p \pm Z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$
Sample Size for Estimating Mean	$n = \frac{Z_{\alpha/2}^2\sigma^2}{e^2}$
Sample Size for the Proportion	$n = \frac{Z^2\pi(1-\pi)}{e^2}$

Fundamentals of Hypothesis Testing One-Sample Test

Test Statistic for Testing a mean, σ known	$Z = \frac{\bar{X}-\mu}{\frac{\sigma}{\sqrt{n}}}$
Test Statistic for Testing a mean, σ unknown	$t = \frac{\bar{X}-\mu}{\frac{s}{\sqrt{n}}}$
Test Statistic for Testing proportion	$Z = \frac{p-\pi}{\sqrt{\frac{\pi(1-\pi)}{n}}}$

Two-Sample Test

Pooled Variance for σ_1 & σ_1 unknown and assumed equal	$S_p^2 = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{(n_1-1)+(n_2-1)}$
Test statistic for σ_1 & σ_1 unknown and assumed equal	$t_{STAT} = \frac{(\overline{X_1}-\overline{X_2})-(\mu_1-\mu_2)}{\sqrt{S_p^2(\frac{1}{n_1}+\frac{1}{n_2})}}$

SAMPLE

Confidence Interval for σ_1 & σ_1 unknown and assumed equal

$$(\overline{X_1} - \overline{X_2}) \pm t_{\alpha/2}\sqrt{S_p^2(\frac{1}{n_1} + \frac{1}{n_2})}$$

Point estimate for the paired difference population mean

$$\overline{D} = \frac{\sum_{i=1}^{n} D_i}{n}$$

Sample Standard Deviation

$$S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \overline{D})^2}{n-1}}$$

Test Statistic for Paired Difference

$$t_{STAT} = \frac{\overline{D} - \mu_D}{\frac{s_D}{\sqrt{n}}}$$

Confidence Interval for Paired Difference

$$\overline{D} \pm t_{\alpha/2}\frac{s_D}{\sqrt{n}}$$

Pooled Estimate for Overall Proportion

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

Two Population Proportion Test Statistic

$$Z_{STAT} = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})(\frac{1}{n_1} + \frac{1}{n_2})}}$$

Two Population Proportion Confidence Interval

$$(p_1 - p_2) \pm Z_{\alpha/2}\sqrt{S_p^2(\frac{p_1}{n_1} + \frac{1}{n_2})}$$

Analysis of Variance (ANOVA)

One-way Total sum of square

$$SST = \sum_{j=1}^{c}\sum_{i=1}^{n_j}(X_{ij} - \bar{\bar{X}})^2$$

One-way Sum of squares among groups

$$SSA = \sum_{j=1}^{c} n_j(\bar{X}_j - \bar{\bar{X}})^2$$

One-way Sum of square within groups

$$SSW = \sum_{j=1}^{c}\sum_{i=1}^{n_j}(X_{ij} - \bar{X}_j)^2$$

One-way Mean Square Among Groups

$$MSA = \frac{SSA}{c-1}$$

One-way Mean Square Within Groups

$$MSW = \frac{SSW}{n-c}$$

SAMPLE

One-way Mean Square Total

$$MST = \frac{SST}{n-1}$$

One-way ANOVA F-test

$$F_{STAT} = \frac{MSA}{MSW}$$

Two-way Total Variation

$$SST = \sum_{i=1}^{r}\sum_{j=1}^{c}\sum_{k=1}^{n'}(X_{ijk} - \bar{\bar{X}})^2$$

Two-way Factor A Variation

$$SSA = cn'\sum_{i=1}^{r}(\bar{X}_{i..} - \bar{\bar{X}})^2$$

Two-way Factor B Variation

$$SSB = rn'\sum_{j=1}^{c}(\bar{X}_{.j.} - \bar{\bar{X}})^2$$

Two-way Interaction Variation

$$SSAB = n'\sum_{i=1}^{r}\sum_{j=1}^{c}(\bar{X}_{ij.} - \bar{X}_{i..} - \bar{X}_{.j.} + \bar{\bar{X}})^2$$

Two-way Sum of Squares Error

$$SSE = \sum_{i=1}^{r}\sum_{j=1}^{c}(X_{ijk} - \bar{X}_{ij.})^2$$

Two-way Grand Mean

$$\bar{\bar{X}} = \frac{\sum_{i=1}^{r}\sum_{j=1}^{c}\sum_{k=1}^{n'}X_{ijk}}{rcn'}$$

Two-way Mean Square Factor A

$$MSA = \frac{SSA}{r-1}$$

Two-way Mean Square Factor B

$$MSB = \frac{SSB}{c-1}$$

Two-way Mean Square Interaction

$$MSAB = \frac{SSAB}{(r-1)(c-1)}$$

Two-way Mean Square Error

$$MSE = \frac{SSE}{rc(n'-1)}$$

Two-way ANOVA F-test Factor A

$$F_{STAT} = \frac{MSA}{MSE}$$

Two-way ANOVA F-test Factor B

$$F_{STAT} = \frac{MSB}{MSE}$$

SAMPLE

Two-way ANOVA F-test Interaction

$$F_{STAT} = \frac{MSAB}{MSE}$$

Common Levels of Confidence

Confidence Level	Confidence Coefficient 1 - α	$Z_{a/2}$ Value
80%	0.80	1.28
90%	0.90	1.645
95%	0.95	1.96
98%	0.98	2.33
99%	0.99	2.58
99.8%	0.998	3.08
99.9%	0.999	3.27

SAMPLE

Z Table (1/2)
Cumulative Standardised Normal Distribution

Cumulative Probabilities										
Z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7518	0.7549
0.7	0.7580	0.7612	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99897	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4.0	0.999968329									
4.5	0.999996602									
5.0	0.999999713									
5.5	0.999999981									
6.0	0.999999999									

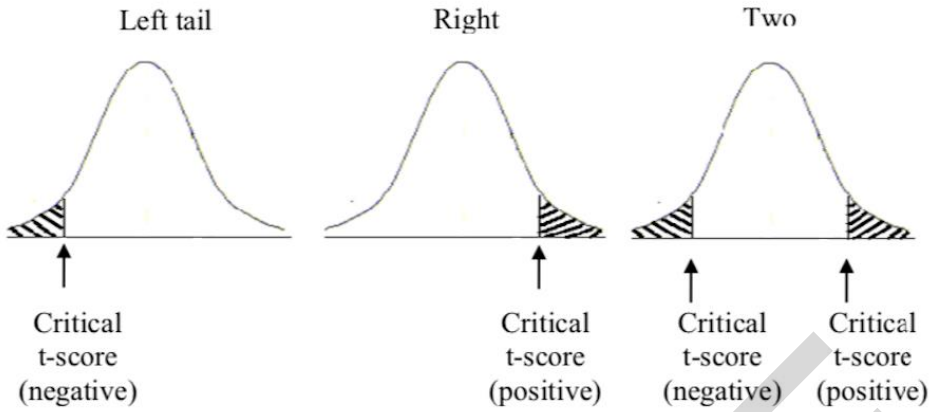
SAMPLE

Z Table (2/2)
Cumulative Standardised Normal Distribution

	Cumulative Probabilities									
Z	**0.00**	**0.01**	**0.02**	**0.03**	**0.04**	**0.05**	**0.06**	**0.07**	**0.08**	**0.09**
−6.0	0.000000001									
−5.5	0.000000019									
−5.0	0.000000287									
−4.5	0.000003398									
−4.0	0.000031671									
−3.9	0.00005	0.00005	0.00004	0.00004	0.00004	0.00004	0.00004	0.00004	0.00003	0.00003
−3.8	0.00007	0.00007	0.00007	0.00006	0.00006	0.00006	0.00006	0.00005	0.00005	0.00005
−3.7	0.00011	0.00010	0.00010	0.00010	0.00009	0.00009	0.00008	0.00008	0.00008	0.00008
−3.6	0.00016	0.00015	0.00015	0.00014	0.00014	0.00013	0.00013	0.00012	0.00012	0.00011
−3.5	0.00023	0.00022	0.00022	0.00021	0.00020	0.00019	0.00019	0.00018	0.00017	0.00017
−3.4	0.00034	0.00032	0.00031	0.00030	0.00029	0.00028	0.00027	0.00026	0.00025	0.00024
−3.3	0.00048	0.00047	0.00045	0.00043	0.00042	0.00040	0.00039	0.00038	0.00036	0.00035
−3.2	0.00069	0.00066	0.00064	0.00062	0.00060	0.00058	0.00056	0.00054	0.00052	0.00050
−3.1	0.00097	0.00094	0.00090	0.00087	0.00084	0.00082	0.00079	0.00076	0.00074	0.00071
−3.0	0.00135	0.00131	0.00126	0.00122	0.00118	0.00114	0.00111	0.00107	0.00103	0.00100
−2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
−2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
−2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
−2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
−2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
−2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
−2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
−2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
−2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
−2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
−1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
−1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
−1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
−1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
−1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
−1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
−1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
−1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
−1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
−1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
−0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
−0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
−0.7	0.2420	0.2388	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
−0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2482	0.2451
−0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
−0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
−0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
−0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
−0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
−0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641

SAMPLE

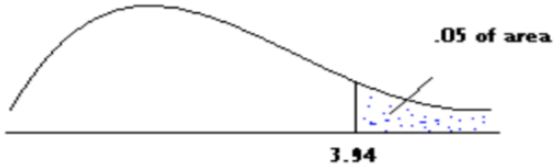
t Table



Degrees of freedom	0.005 (one tail) 0.01 (two tails)	0.01 (one tail) 0.02 (two tails)	0.025 (one tail) 0.05 (two tails)	0.05 (one tail) 0.10 (two tails)
1	63.657	31.821	12.706	6.314
2	9.925	6.965	4.303	2.920
3	5.841	4.541	3.182	2.353
4	4.604	3.747	2.776	2.132
5	4.032	3.365	2.571	2.015
6	3.707	3.143	2.447	1.943
7	3.500	2.998	2.365	1.895
8	3.355	2.896	2.306	1.860
9	3.250	2.821	2.262	1.833
10	3.169	2.764	2.228	1.812
11	3.106	2.718	2.201	1.796
12	3.054	2.681	2.179	1.782
13	3.012	2.650	2.160	1.771
14	2.977	2.625	2.145	1.761
15	2.947	2.602	2.132	1.753
16	2.921	2.584	2.120	1.746
17	2.898	2.567	2.110	1.740
18	2.878	2.552	2.101	1.734
19	2.861	2.540	2.093	1.729
20	2.845	2.528	2.086	1.725
21	2.831	2.518	2.080	1.721
22	2.819	2.508	2.074	1.717
23	2.807	2.500	2.069	1.714
24	2.797	2.492	2.064	1.711
25	2.787	2.485	2.060	1.708
26	2.779	2.479	2.056	1.706
27	2.771	2.473	2.052	1.703
28	2.763	2.467	2.048	1.701
29	2.756	2.462	2.045	1.699
Large (Z)	**2.575**	**2.327**	**1.960**	**1.645**

SAMPLE

F Table (1/2)
F Distribution with 0.05 of the area in the Right Tail



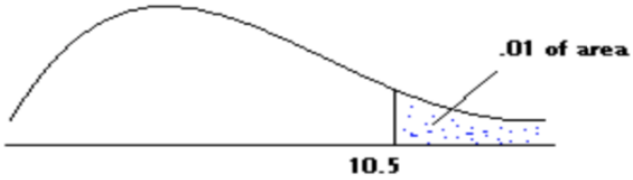
EXAMPLE: For a test of a significance of .05 where we have 15 degrees of freedom for the numerator and 6 degrees of freedom for the denominator, the appropriate F value is found by looking under the 15 degrees of freedom column and proceeding down to the 6 degrees of freedom row; there we find the appropriate F value to be 3.94

Degrees of freedom for numerator

Degrees of freedom for denominator	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120	x
1	161	200	216	225	230	234	237	239	241	242	244	246	248	249	250	251	252	253	254
2	18.5	19.0	19.2	19.3	19.3	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.4	19.5	19.5	19.5	19.5	19.5	19.5
3	10.1	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81	8.79	8.74	8.7	8.66	8.64	8.62	8.59	8.57	8.55	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00	5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.69	5.66	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77	4.74	4.68	4.62	4.56	4.53	4.50	4.46	4.43	4.40	4.37
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10	4.06	4.00	3.94	3.87	3.84	3.81	3.77	3.74	3.70	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68	3.64	3.57	3.51	3.44	3.41	3.38	3.34	3.30	3.27	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39	3.35	3.28	3.22	3.15	3.12	3.08	3.04	3.01	2.97	2.93
9	5.12	4.30	3.86	3.63	3.48	3.37	3.29	3.23	3.18	3.14	3.07	3.01	2.94	2.90	2.86	2.83	2.79	2.75	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02	2.98	2.91	2.85	2.77	2.74	2.70	2.66	2.62	2.58	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90	2.85	2.79	2.72	2.65	2.61	2.57	2.53	2.49	2.45	2.40
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80	2.75	2.69	2.62	2.54	2.51	2.47	2.43	2.38	2.34	2.30
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71	2.67	2.60	2.53	2.46	2.42	2.38	2.34	2.30	2.25	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65	2.60	2.53	2.46	2.39	2.35	2.31	2.27	2.22	2.18	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59	2.54	2.48	2.40	2.33	2.29	2.25	2.20	2.16	2.11	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54	2.49	2.42	2.35	2.28	2.24	2.19	2.15	2.11	2.06	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49	2.45	2.38	2.31	2.23	2.19	2.15	2.10	2.06	2.01	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46	2.41	2.34	2.27	2.19	2.15	2.11	2.06	2.02	1.97	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42	2.38	2.31	2.23	2.16	2.11	2.07	2.03	1.98	1.93	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39	2.35	2.28	2.20	2.12	2.08	2.04	1.99	1.95	1.90	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37	2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.92	1.87	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34	2.30	2.23	2.15	2.07	2.03	1.98	1.94	1.89	1.84	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32	2.27	2.20	2.13	2.05	2.01	1.96	1.91	1.86	1.81	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30	2.25	2.18	2.11	2.03	1.98	1.94	1.89	1.84	1.79	1.73
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28	2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.82	1.77	1.71
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21	2.16	2.09	2.01	1.93	1.89	1.84	1.79	1.74	1.68	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12	2.08	2.00	1.92	1.84	1.79	1.74	1.69	1.64	1.58	1.51
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04	1.99	1.92	1.84	1.75	1.70	1.65	1.59	1.53	1.47	1.39
120	3.92	3.07	2.68	2.45	2.29	2.18	2.09	2.02	1.96	1.91	1.83	1.75	1.66	1.61	1.55	1.50	1.43	1.35	1.25
x	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88	1.83	1.75	1.67	4.57	15.20	1.46	1.39	1.32	1.22	1.00

SAMPLE

F Table (2/2)
F Distribution with 0.01 of the Area in the Right Tail



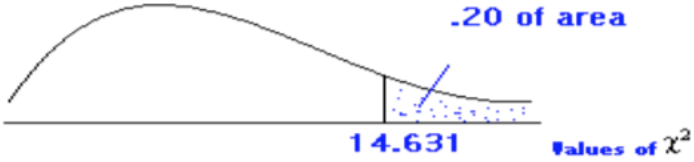
EXAMPLE: For a test of a significance of .01 where we have 7 degrees of freedom for the numerator and 5 degrees of freedom for the denominator, the appropriate F value is found by looking under the 7 degrees of freedom column and proceeding down to the 5 degrees of freedom row; there we find the appropriate F value to be 10.5

Degrees of freedom for numerator

Degrees of freedom for denominator	1	2	3	4	5	6	7	8	9	10	12	15	20	24	30	40	60	120
1	4.052	5.000	5.403	5.625	5.764	5.859	5.928	5.982	6.023	6.056	6.106	6.157	6.209	6.235	6.261	6.287	6.313	6.339
2	98.5	99	99.2	99.2	99.3	99.3	99.4	99.4	99.4	99.4	99.4	99.4	99.4	99.5	99.5	99.5	99.5	99.5
3	34.1	30.8	29.5	28.7	28.2	27.9	27.7	27.5	27.3	27.2	27.1	26.9	26.7	26.6	26.5	26.4	26.3	26.2
4	21.2	18.0	16.7	16	15.5	15.2	15.0	14.8	17.7	14.5	14.4	14.2	14.0	13.9	13.8	13.7	13.7	13.6
5	16.3	13.3	12.1	11.4	11.0	10.7	10.5	10.3	10.2	10.1	9.89	9.72	9.55	9.47	9.38	9.29	9.20	9.11
6	13.7	10.9	9.78	9.15	8.75	8.47	8.26	8.10	7.98	7.87	7.72	7.56	7.40	7.31	7.23	7.14	7.06	6.97
7	12.2	9.55	8.45	7.85	7.46	7.19	6.99	6.84	6.72	6.62	6.47	6.31	6.16	6.07	5.99	5.91	5.82	5.74
8	11.3	8.65	7.59	7.01	6.63	6.37	6.18	6.03	5.91	5.81	5.67	5.52	5.36	5.28	5.20	5.12	5.03	4.95
9	10.6	8.02	6.99	6.42	6.06	5.80	5.61	5.47	5.35	5.26	5.11	4.96	4.81	4.73	4.65	4.57	4.48	4.40
10	10.0	7.56	6.55	5.99	5.64	5.39	5.20	5.06	4.94	4.85	4.71	4.56	4.41	4.33	4.25	4.17	4.08	4.00
11	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.74	4.63	4.54	4.40	4.25	4.10	4.02	3.94	3.86	3.78	3.69
12	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.50	4.39	4.30	4.16	4.01	3.86	3.78	3.70	3.62	3.54	3.45
13	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.30	4.19	4.10	3.96	3.82	3.66	3.59	3.51	3.43	3.34	3.25
14	8.86	6.51	5.56	5.04	4.70	4.46	4.28	4.14	4.03	3.94	3.80	3.66	3.51	3.43	3.35	3.27	3.18	3.09
15	8.68	6.36	5.42	4.89	4.56	4.32	4.14	4.00	3.89	3.80	3.67	3.52	3.37	3.29	3.21	3.13	3.05	2.96
16	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.89	3.78	3.69	3.55	3.41	3.26	3.18	3.10	3.02	2.93	2.84
17	8.40	6.11	5.19	4.67	4.34	4.10	3.93	3.79	3.68	3.59	3.46	3.31	3.16	3.08	3.00	2.92	2.83	2.75
18	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71	3.60	3.51	3.37	3.23	3.08	3.00	2.92	2.84	2.75	2.66
19	8.19	5.93	5.01	4.50	4.17	3.94	3.77	3.63	3.52	3.43	3.30	3.15	3.00	2.92	2.84	2.76	2.67	2.58
20	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56	3.46	3.37	3.23	3.09	2.94	2.86	2.78	2.69	2.61	2.52
21	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51	3.40	3.31	3.17	3.03	2.88	2.80	2.72	2.64	2.55	2.46
22	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45	3.35	3.26	3.12	2.98	2.83	2.75	2.67	2.58	2.50	2.40
23	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41	3.30	3.21	3.07	2.93	2.78	2.70	2.62	2.54	2.45	2.35
24	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36	3.26	3.17	3.03	2.89	2.74	2.66	2.58	2.49	2.40	2.31
25	7.77	5.57	4.68	4.18	3.86	3.63	3.46	3.32	3.22	3.13	2.99	2.85	2.70	2.62	2.53	2.45	2.36	2.27
30	7.56	5.39	4.51	4.02	3.70	3.47	3.30	3.17	3.07	2.98	2.84	2.70	2.55	2.47	2.39	2.30	2.21	2.11
40	7.31	5.18	4.31	3.83	3.51	3.29	3.12	2.99	2.89	2.80	2.66	2.52	2.37	2.29	2.20	2.11	2.02	19.2
60	7.08	4.98	4.13	3.65	3.34	3.12	2.95	2.82	2.72	2.63	2.50	2.35	2.20	2.12	2.03	1.94	1.84	1.73
120	6.85	4.79	3.95	3.48	3.17	2.96	2.79	2.66	2.56	2.47	2.34	2.19	2.03	1.95	1.86	1.76	1.66	1.53
x	6.63	4.61	3.78	3.32	3.02	2.80	2.64	2.51	2.41	2.32	2.18	2.04	1.88	1.79	1.70	1.59	1.47	1.32

SAMPLE

Area in the Right Tail of a Chi-square Distribution



EXAMPLE: In a chi-square distribution with 11 degrees of freedom. If we want to find the appropriate chi-square value for 0.05 of the area under the curve (the area in the right tail) we look under the 0.05 column in the table and proceed down to the 11 degrees of freedom row; the appropriate chi-square value there is 19.675.

Area in right tail

Degrees of freedom	.05	.01
1	3.841	6.635
2	5.991	9.210
3	7.815	11.345
4	9.488	13.277
5	11.070	15.086
6	12.592	16.812
7	14.067	18.475
8	15.507	20.090
9	16.919	21.666
10	18.307	23.209
11	19.675	24.725
12	21.026	26.217
13	22.362	27.688
14	23.685	29.141
15	24.996	30.578
16	26.296	32.000
17	27.587	33.409
18	28.869	34.805
19	30.144	36.191
20	31.410	37.566
21	32.671	38.932
22	33.924	40.289
23	35.172	41.638
24	36.415	42.980
25	37.652	44.314
26	38.885	45.642
27	40.113	46.963
28	41.337	48.278
29	42.557	49.588
30	43.773	50.892

SAMPLE